

TEST PLAN DOCUMENT

Project: Automated Independent Validation System (Java)

Application Under Test: Notes App – ExpandTesting

Version: 1.0

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Role: Quality Engineer – Automation

Date: 21-05-2026

1. Introduction

This Test Plan outlines the testing strategy, scope, resources, schedule, environments, risks, and deliverables for validating the Notes Application using both Manual Testing and Automated Testing (Selenium + Java + RestAssured).

The goal of testing is to ensure functional accuracy, UI correctness, API reliability, and synchronization between UI and API layers as required by the capstone evaluation standards.

2. Objective

The primary objectives of testing include:

- Validate functional requirements of the Notes App through UI and API testing.
- Ensure synchronization between UI-created data and API responses.
- Identify defects early and ensure product stability.
- Provide confidence that the system works correctly using end-to-end hybrid validation.
- Prepare the foundation for automation with a clear manual baseline.

3. Scope

3.1 In-Scope

- UI Testing:
 - Login
 - Create Note
 - Edit Note
 - Delete Note
 - Category-wise listing
 - DOM update validations

- API Testing:
 - Authentication
 - POST /notes
 - GET /notes
 - DELETE /notes/{id}
 - Negative API validations (invalid token, missing fields, etc.)
- End-to-End Hybrid Testing:
 - UI → API Data Consistency
 - API → UI Sync
- Performance checks:
 - API response time <2 seconds
 - UI load verification
- Validation of both functional and non-functional requirements.

3.2 Out-of-Scope

- Load Testing
- Security Testing
- Database Testing
- Backend architecture validation
- Performance benchmarking / JMeter tests
- Cross-browser testing beyond Chrome unless requested

4. Test Strategy

4.1 Manual Testing Strategy

- Explore UI and API functionality to understand user flows.
- Prepare Test Scenarios and Test Cases based on real application behavior.
- Execute manual test cases for baseline validation.
- Log defects with steps, screenshots, and severity.

4.2 Automation Testing Strategy

Automation will be done after manual sign-off.

UI Automation (Selenium + Java)

- Framework: POM
- Selenium WebDriver + WebDriverManager
- TestNG for execution
- Explicit waits, custom utilities
- ThreadLocal driver for parallel tests

API Automation (RestAssured)

- Base request specs
- Schema validation
- Response time checks
- Retry logic

Hybrid Automation

- Create note via UI → validate via API
- Delete via API → verify removal in UI

Reports

- Allure
- HTML Reports

5. Test Deliverables

Functional QA Deliverables

- Test Plan
- Test Scenarios
- Test Cases

- Manual Test Execution Report
- Defect Report
- RTM (Requirement Traceability Matrix)

Automation QA Deliverables

- Automation Framework (UI + API + Hybrid)
- Test Execution Screenshots & Logs
- Allure Report
- HTML Report
- CI/CD Pipeline (Jenkins)
- Parallel Suite XML
- AI-driven enhancements (self-healing locators, retry logic)

6. Test Environment

UI URL:

<https://practice.expandtesting.com/notes/app>

API Swagger URL:

<https://practice.expandtesting.com/notes/api/api-docs/>

Environment Components

Component	Details
Browser	Chrome (latest)
Testing Tools	Selenium, TestNG, RestAssured
IDE	IntelliJ IDEA / Eclipse
Reporting	Allure, Extent
CI/CD	Jenkins
OS	Windows / Linux

7. Roles & Responsibilities

Role	Responsibilities
QA Engineer	Create Test Plan, Scenarios, Cases, Manual Execution, Defect Logging, Automation Framework, Reports
Mentor / Reviewer	Review deliverables, provide feedback
Dev Team	Support defect fixes

8. Testing Schedule

Activity	Start Date	End Date
Requirement Analysis	Day 1	Day 1
Manual Test Design	Day 1	Day 1
Manual Execution	Day 1	Day 1
Automation Framework Setup	Day 2	Day 2
UI Automation	Day 2	Day 2
API Automation	Day 3	Day 3
Hybrid Automation	Day 4	Day 4
Reporting + Submission	Day 5	Day 5

9. Entry & Exit Criteria

Entry Criteria

- Requirements understood
- Application accessible (UI + API)
- Test data prepared

Exit Criteria

- 100% test cases executed
- All major defects closed or accepted
- Automation scripts executed and stable
- Reports generated

10. Risk & Mitigation

Risk	Impact	Probability	Mitigation
API downtime	Blocks testing	Medium	Test during stable hours
UI locator changes	Script failures	High	Implement self-healing locators
Execution delays	Missed deadlines	Medium	Parallel testing via TestNG
Flaky tests	Unreliable results	Medium	Retry logic + stable waits

11. Tools & Technologies

- Selenium WebDriver
- Java 17+
- TestNG
- RestAssured
- Allure Report
- WebDriverManager
- Jenkins
- Gson / Jackson
- Maven

12. Approval

Name	Role	Status
Ajinkya Sandip Ronghe	QA Engineer	Approved
Mr/Ms XYZ	Reviewer	Pending